

Andrew Clark

Senior Software Engineer

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PROFILE

Senior Backend Engineer with **14+ years** of experience architecting **high-scale distributed systems** (50B+ daily requests, \$2M+/month platforms). Proven expertise in:

- **Cloud-Native Backends:** Designed **gRPC microservices** (22K RPS, <30ms P99) and **GraphQL federations** for Fortune 500 clients.
- **Performance Optimization:** Reduced latency by **40-75%** across Python/Go/Node.js systems via **Redis, Protobuf, and PostgreSQL tuning**.
- **Cost Efficiency:** Saved **\$300K+/year** in AWS costs through **Kubernetes spot instances, serverless architectures, and Terraform automation**.
- **Full SDLC Leadership:** Scaled teams from zero to **10+ engineers**, mentoring junior developers and enforcing **SLA-driven development**.
- **Enterprise Security:** Implemented **mTLS, OAuth2.0, and HIPAA-compliant** data pipelines with zero breaches.

Core Stack: Golang, Python, Node.js, AWS (EKS/Lambda/DynamoDB), gRPC, GraphQL, Kafka, PostgreSQL, Kubernetes.

EDUCATION

Bachelor's in Computer Science
Northeastern University

09/2007 – 11/2010
Boston,
Massachusetts

SKILLS

Languages & Frameworks

- **Golang** (Gin, Fiber, gRPC, Protobuf)
- **Python** (Django, FastAPI, Flask, Celery)
- **Node.js/TypeScript/Javascript** (Express, NestJS)

Cloud & DevOps

- **AWS:** EKS/ECS, Lambda, S3, DynamoDB, RDS, Cognito, CDK, CloudFormation
- **Infra-as-Code:** Terraform, Pulumi
- **CI/CD:** GitHub Actions, ArgoCD, Jenkins
- **Containerization:** Docker, Kubernetes (Helm, Operators)
- **Monitoring:** Prometheus, Grafana, OpenTelemetry, ELK

Databases & Data Engineering

- **SQL:** PostgreSQL (Partitioning, PL/pgSQL), MySQL
- **NoSQL:** DynamoDB (Single-Table Design), MongoDB, Cassandra
- **Event Streaming:** Kafka, RabbitMQ, AWS SQS/SNS
- **Cache:** Redis (Cluster, Lua Scripting), Memcached

API & Architecture

- **API Protocols:** REST (10+ yrs), **GraphQL** (Apollo Federation), **gRPC** (Load Balancing, mTLS)
- **Architectures:** Microservices, Event-Driven, CQRS, Serverless
- **Security:** OAuth2.0, JWT, AWS Cognito, mTLS, HIPAA/GDPR

Performance Optimization

- High-Concurrency Systems (50K+ WebSocket connections)
- Query Optimization (PostgreSQL, DynamoDB)
- Latency Reduction (Protobuf, Connection Pooling)

PROFESSIONAL EXPERIENCE

Senior Backend Engineer

Sound Physicians

10/2020 – 04/2025

United States

- Designed and implemented a **Golang-based gRPC microservices architecture** handling **22,000 RPS** with <30ms P99 latency for a high-frequency trading platform, using **connection pooling, circuit breakers, and binary Protobuf serialization** to reduce network overhead by 40% compared to REST/JSON.
- Led migration of **12 legacy Python/Node.js monoliths** to Kubernetes (EKS) using **Terraform and ArgoCD**, achieving **zero downtime deployments** and reducing AWS costs by \$280K/year through spot instance optimization and auto-scaling policies.
- Built a **real-time GraphQL federation layer** (Apollo Router) unifying 8 disparate APIs into a single endpoint, reducing client-side data fetching complexity by 65% and improving mobile app performance by 33% through query batching and Redis caching.
- Developed an **event-driven order processing system** with **Kafka and DynamoDB Streams**, ensuring exactly-once semantics for 5M+ daily transactions in a financial settlement platform.
- Implemented **mTLS authentication** across all internal gRPC services using **Istio and cert-manager**, eliminating man-in-the-middle attack vectors.
- Created a **Prometheus/Grafana/OpenTelemetry observability stack** monitoring 150+ microservices, reducing incident MTTR from 47 minutes to 8 minutes through automated anomaly detection.
- Mentored 9 engineers in **distributed systems design patterns** (CQRS, Saga, Event Sourcing), resulting in a 70% reduction in production incidents caused by race conditions.

Senior Full Stack Developer

StratusLogic Solutions

03/2017 – 09/2020

- Architected a **Node.js/TypeScript** backend for a healthcare analytics platform processing **18TB of EHR data daily**, using **PostgreSQL partitioning** to reduce query times from 12s to 300ms for complex joins across 50M+ patient records.
- Built a **serverless data pipeline** with **AWS Lambda, S3, and Glue** that transformed and loaded 3M+ clinical trial records nightly into a **DynamoDB single-table design**, cutting ETL costs by 60% compared to traditional EC2-based approaches.
- Designed a **GraphQL API with Apollo Federation** serving 200+ frontend applications, implementing **query complexity analysis** to prevent denial-of-service attacks and **persisted queries** to reduce parsing overhead.
- Developed a **Python-based ML feature store** using **Redis and Protobuf** that reduced model training times by 45% through efficient feature vector serialization.
- Spearheaded migration from **MongoDB to PostgreSQL** for ACID compliance, implementing **row-level security policies** to meet HIPAA requirements for 2M+ patient records.
- Optimized a **WebSocket-based real-time notification system** handling 250K+ concurrent connections using **Elixir (for hot paths) and Go (for background workers)**.

Backend engineer

VertexNova Systems

03/2015 – 01/2017

United States

- Created a **Python/Django REST API** for an ad-tech platform processing **50B+ bid requests daily**, leveraging **Redis Cluster and Lua scripting** to achieve 1ms response times for auction decisions.
- Implemented a **Kubernetes-based CI/CD system** using **Jenkins and Spinnaker** that reduced deployment times from 2 hours to 7 minutes across 3 global regions.

- Built a **gRPC inter-service communication layer** with **retry policies, deadline propagation, and distributed tracing** (Jaeger), improving system reliability during AWS AZ failures.
- Designed a **multi-tenant SaaS architecture** using **PostgreSQL schemas and Django ORM hooks**, isolating data for 1,200+ enterprise clients while maintaining single-codebase efficiency.
- Developed a **custom OLAP query engine** in **Go** that aggregated analytics data 10x faster than the previous Snowflake-based solution by using **columnar storage and SIMD optimizations**.

Software Engineer

TitanForge Labs

01/2011 – 01/2015

Boston, MA,

United States

- Built the **first version of an e-commerce platform** handling \$2M/month in transactions using **PHP/Laravel**, later migrating to **Python/Flask** to improve performance by 400%.
- Implemented a **WebSocket-based inventory management system** that reduced overselling incidents by 90% by synchronizing product stock across 20+ fulfillment centers in real-time.
- Created a **batch processing framework** using **Celery and RabbitMQ** that generated 10,000+ personalized PDF invoices nightly with **3D barcodes for logistics tracking**.
- Designed a **MySQL sharding strategy** that horizontally scaled the product catalog to 50M+ SKUs while maintaining <100ms search latency through **denormalized Elasticsearch indices**.
- Developed a **custom TCP protocol** for point-of-sale devices that reduced credit card transaction times from 8s to 1.2s in areas with poor connectivity.